

CS201 Midterm 1 Part A

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I. QUESTION (ALGORITHM ANALYSIS)

```
1 int algorithm1(int N){
2     int i , j , k, sum = 0;
3     for (i = 1; i <= N - 3 * i; i++)
4         sum++;
5     for (i = 1; i <= N; i *= 2)
6         for (j = N; j >= 1; j /= 2)
7             sum++;
8     for (i = 1; i <= N; i++)
9         for (j = i + 1; j <= i * i; j++)
10            sum++;
11     return sum;
12 }
```

What does the function algorithm1 return in terms of N? Assume that N is a power of 2.

II. QUESTION (ALGORITHM ANALYSIS)

```
1 int algorithm2(int N){
2     if (N == 1)
3         return 0;
4     int sum = 0;
5     for (int i = N; i > 0; i--)
6         for (int j = 0; j < N; j++)
7             sum++;
8     for (int i = 0; i < 4; i++)
9         for (int j = 0; j < 2; j++)
10            sum += algorithm2(N / 4);
11     return sum;
12 }
```

What is the time complexity of algorithm2? You must construct the recurrence equation and solve it without using the master theorem.